Anukool Yadav

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Skills

- **Languages**: C++, C, Python, SQL
- **Frameworks**: Tensorflow, PyTorch, Keras, Scikit-Learn, Pandas, NumPy
- **Tools/Platforms**: Microsoft Excel, Google Sheets, MySQL, PostSQL, Unity, Blender
- **Core Skills**: Data Collection, Data Validation, Data Cleaning, Report Preparation, Data Analysis, Online Research

Internship

- **Cipher Schools** — Jan 2025 - Feb 2025
  Training
  - **Developed** Interactive learning modules and coding challenges to enhance understanding of Data Structures and Algorithms (DSA) and Machine Learning.
  - **Engineered** Problem-solving exercises that strengthened algorithmic thinking and coding proficiency.
  - **Authored** well-documented code and debugged critical issues to maintain system efficiency.

Projects

- **Air Quality Monitoring & PM2.5 Prediction Using Deep Learning** — Oct 2025 - Nov 2025
  - Developed and compared CNN, LSTM, and CNN-LSTM hybrid models for PM2.5 forecasting using large-scale spatio-temporal air quality data, achieving ~22% lower RMSE and R^2 up to 0.37 over baseline CNN models.
  - Implemented an end-to-end data preprocessing and feature engineering pipeline (normalization, temporal lag features, rolling statistics), enabling stable training and improved generalization across 6,000+ time-series samples.
  - Performed comprehensive model evaluation and error analysis (Actual vs Predicted plots, MAE/RMSE/R square ≤ comparison, residual distribution, temporal error drift), identifying systematic under-prediction at high pollution levels and guiding model optimization.

  Tech: Python, TensorFlow, Keras, CNN, LSTM, CNN-LSTM, Scikit-learn, Matplotlib

- **Crop Maturity Stage Detection from Field Images** — Feb 2026 - April 2026
  - Developed a deep learning-based crop maturity classification system using VGG-16, ResNet-50, and DenseNet-121 to identify Early, Mid, and Mature crop growth stages from field images.
  - Built an end-to-end image processing pipeline including image resizing, normalization, augmentation, and feature extraction, improving model robustness under varying environmental conditions..
  - Trained and evaluated multiple CNN architectures on a dataset of 3,300+ agricultural images, achieving 89.8% accuracy with VGG-16 as the best-performing model.

  Tech: Python, PyTorch, Scikit-Learn, Deep Learning, Computer Vision, CNN, VGG-16, ResNet-50, DenseNet-12

Certificates

- Azure AI Fundamentals by Microsoft
- Mastering Data Structures & Algorithm using C & C++ by Udemy
- Unity Game Development Advanced
- Fundamentals of AI Agents Using RAG and LangChain

Achievements

- **Solved 275+ problems on Leetcode**:
  Earned 6 achivement badges.
- **Earned HackerRank Badges**:
  Having C++(5 stars), Python(2 stars), SQL(3 stars).
- **Published a Research Paper at ICIRDST-2025:**
  International Conference on Innovations and Research Directions in Science and Technology.

Education

- **Lovely Professional University** — Punjab, India
  Bachelor of Technology - Computer Science and Engineering; ***CGPA:*** *7.5* — *Since August 2022*
- **Raath International School** — Alwar, Rajasthan
  Intermediate; ***Percentage:*** *83.2%* — *April 2019 - March 2020*
- **Mohan Lal Dayal Vinay Mandir** — Neemrana, Rajasthan
  Matriculation; ***Percentage:*** *81.6%* — *April 2017 - March 2018*